Stormwatch (Coastal / Storm) Map Design

Executive Summary

Stormwatch is a violent **stormridden coastal battlefield** declared around **weather disruption. visibility loss. and sensor deoradation**. The terrain features iaaeed cliffs. storm-battered beaches. flooded roads. industrial seawalls. offshore turbines. and emerecency shelters. The mao emphasizes Obsidian Protocol's autonomous adaptation to extreme weather. sensor-fusion under interference. comms-relay survival. and loaistics under raaidly shiftine environmental conditions. In AR (Nreal Liaht)Ori. the mao anchors at ~4–5 m viewine distance. coverine a 10×10 m ohvsical area representine ~1 km² of storm-struck coastline. Kev features include lathtnine-damaaed infrastructure. storm suree zones. flooded combat routes. and multi-domain operations under severe visibility constraints.

Design Goals & Gameplay

Extreme Weather Combat
Stormwatch prioritizes combat under heavy rain, fog, lightning, and wind. Visibility constantly shifts, affecting targeting, movement, and sensor reliability.
Autonomy & Weather Adaptation
Autonomous AI reroutes around flooded roads, avoids lightning-exposed high ground, and uses sheltered structures for cover. Units adapt to rapidly changing visibility.
Sensor & Information Warfare
Rain scatter, fog density, lightning interference, and turbulent air distort sensors. Relay towers on cliffs and turbines restore clarity. Destroying relays plunges entire sectors into sensor blackout.
Logistics & Resources
Storm-damaged docks, emergency shelters, and industrial turbines serve as resupply nodes. Supply convoys must navigate flooded roads and exposed coastal paths.
Deployment & Progress
A fixed deolovment budaet (~10.000 aoints) forces olavers to choose between storm-resistant units. fast recon drones. or heavy armor. Blue deolovs from the southwest storm shelter: Red deolovs from the northeast cliffside command.

Spatial Constraints (Nreal AR)

Device Capabilities
Nreal Light/Air: ~52° diagonal FOV, 1920×1080 per-eye resolution. Optimal viewing distance ~4–5 m.
Playable Area & Scaling
10×10 m physical area = 1 km² storm-struck coastal zone (1:100 scale). Designed for tabletop AR.
Tracking & Occlusion
Inside-out tracking detects horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; assume minimal real-world occlusion.

Layout Overview

Coordinate System
1000×1000 grid (1 unit = 1 m). Origin (0,0) = southwest corner.
Key Locations
Blue HQ (Storm Shelter Command) — (0,0). 140×140 m Reinforced emergency shelter with storm-resistant vehicles.
Red HQ (Cliffside Command Tower) — (950,950). 140×140 m slab-around command tower with radar mast and AA emplacements.
Central Floodplain — (500,450). 250×250 m Flooded roads. debris. abandoned vehicles. low visibility.
Industrial Turbine Field — (250,800). 120×120 m Wind turbines. generators. lightning-damaged infrastructure.
Storm-Damaged Dockyard — (800,300). 150×150 m Broken piers. boats. cranes. flooded warehouses.
Storm Surge Channel — (0,450 → 400,900) Rapid water flow: dangerous crossing zones.
Sheltered Coastal Road — (500,0 → 450,550) Partially covered route: reduced wind exposure.
Cliff Relay Ridge — (650,150). +260 m Relay tower with long-range sensor coverage.
Foreshore Forest Patch — (150,650). +240 m Dense flor: stealth and ambush zone.
Neutral Forward Outpost — (500,950) Drone resupply zone.

Chokepoints & Sightlines

Storm Surge Channel
Crossings are hazardous; naval units can support ground forces.
Floodplain
Low visibility; ideal for ambushes and close-range combat.
Dockyard Ruins
Mixed-range combat with interior flanking routes.
Cliff Relay Ridge
Long-range sightlines but exposed to lightning.
Sightlines
Relay Ridge provides long-range sensor coverage (~300 m). Fog, rain, and lightning create dynamic fog-of-war pockets.

Deployment Zones

Blue Deployment: SW storm shelter + rear cells point.
Red Deployment: NE cliffside command tower + AA emplacements
Neutral Forward Zone: Outpost at (500,950) for drone drops.

Assets & Environment

Terrain
Cliffs, beaches, flooded plains, storm surge channels, coastal roads.
Vegetation
Sparse coastal shrubs, wetland grasses, wind-bent trees.
Buildings & Props
Military: Bunkers, AA guns, radar dishes, watchtowers. Industrial: Turbines, generators, cranes, pipelines. Logistics: Boats, trucks, crates, fuel drums.
Effects
Heavy rain, fog, lightning flashes, wind gusts, water spray.
Audio
Thunder, waves, wind, machinery, radio chatter.

LOD & Budget

Asset Type	Count	Tri/Item	Total Tris
Cliffs/Coa	3	200k	600k
Floodplain	4	80k	320k
Buildings	15	50k	750k
Roads/Pa	4	30k	120k
Vegetatio	150	2k	300k
Props	50	10k	500k
Total	—	—	**~2.59M**

Texture Budget
2048² for terrain/buildings, 1024² for props. DXT1/5 compression. Target ≤100 MB.

Unreal / Nreal Integration

Engine & Plugins
Unreal AR template + Nreal SDK (NREAL). ARKit/ARCore enabled.
Rendering
Forward renderer, stationary lights, baked GI, aggressive culling.
Anchors
Single spatial anchor for entire map.
Interaction
Phone pointer for pan/zoom; HUD as screen-space widgets.
Multiplayer
Client-server architecture, offload AI/physics to server.

Implementation Plan & Testing

Milestones
1. Prototvee Layout — Month 1 Block-out cliffs. floodplain. dockyard. AR anchor test.
2. Core Svstems — Month 2 Roadblock. weather adaptive autonomy behaviors
3. AR Interaction — Month 3 Anchors. plane detection. basic interactivity.
4. Content Fill — Month 4-5 Final models. textures. lighting. audio.
5. Gameplay & Polish — Month 6-7 Objectives. balance. VFX/sfx. fog tuning.
6. Testing & QA — Month 8 Multiplayer stress test. sensor validation.

Testing Plan

Autonomy
Verify weather-adaptive logic, floodplain navigation, storm-surge avoidance.
Comms/Sensors
Test relay destruction, fog-of-war updates, LOS checks under weather.
Performance
≥60 Hz AR render on target device.
QA Checklist
Navigation, weather response, deployment budget enforcement, anchor drift, sensor accuracy.